A0 distilled from B1 NER

CS781 Phase 2 legal named entity recognition | source-audited report

Task and evidence

Predict exact original-character entity spans in 12 legal categories. The source archive holds the actual code, configuration, run record, selected weights, and prediction ZIP. This report was produced after training from those records; it is not an executed historical notebook.

Architecture

A 12-layer causal A0 student is fully tuned. The completed B1 LoRA NER model supplies temperature-2 teacher probabilities on training windows only; the loss mixes hard-label cross-entropy and teacher KL equally.

Path	Legal text -> A0 tokenizer -> 12-layer A0 student -> B1 teacher on train
Output	CE + temperature-2 KL -> Character BiGRU -> Span decoder

Training and validation

Method: Supervised + teacher KL. This export's archived held-out strict macro-F1 is 0.409579 on the 12-document local validation split. Its selected checkpoint and prediction ZIP hashes are in evidence/run_summary.json. The official test set has 35 document IDs; no hidden Codabench score is established by this archive.

Implementation map

Key original files: lner/models.py, lner/train.py, backbones/cs781/models/a.py. The companion notebook explains the tensor route and reads archived evidence without presenting invented execution outputs.

Reproducibility boundary

The package includes exact source snapshots and small run records, plus the archived prediction ZIP. Large model weights, tokenizers, and course data stay in the separate complete archive. Original machine paths in config files need remapping for a new environment. The consensus rule was selected on validation; 100M/200M D222 exports were completed after the original deadline where marked.

Interpretation

Model variants change either pretrained backbone, parameter-update method, NER head, or pretrained checkpoint. A score difference does not isolate one factor unless all other choices match. The local held-out set is small, so its ranking should not be described as a hidden-test result.

Recorded model and adaptation

The source and run records together identify the following settings. Counts include the Phase 2 NER head. The method changes which backbone parameters are updated; it does not change the original Phase 1 tokenizer during this run.

Field	Archived value
Base family	A0
Adaptation	Supervised + teacher KL
Total parameters	110844121
Trainable parameters	110844121
Backbone LR	2e-05
Head LR	0.0003
Pretrained SHA-256	1e8aca9800b87e952ed8…
Selected NER SHA-256	2e605bafac82bfab0bc6…

Training signal and exact-span evaluation

For the common NER runner, the character tagger minimizes weighted cross-entropy over 25 BIO categories; A0 also matches its verified B1 teacher with temperature-2 KL. D222 adds CRF negative log-likelihood and a weighted boundary objective. HeaderCRF combines cross-entropy, CRF, and boundary losses. The validation scorer counts a prediction as correct only when the start, end, and label all match the gold span; macro-F1 averages the 12 per-label F1 values equally.

Observed optimization trace

These five entries are sampled from the archived training log; loss is a per-batch training value, not held-out macro-F1. Its decline alone cannot establish improved entity extraction.

Optimizer step	Epoch	Logged batch loss
1	1	3.1408
500	3	0.3538
1000	6	0.2802
1500	9	0.3046
2000	12	0.3138

Label-level held-out scores

The following F1 values come from the archived selected validation prediction. Low-frequency labels are sensitive to a few documents, so read them beside their support and error counts in evidence/selection.json.

Entity label	F1	TP	FP	FN
APP	0.097	9	68	100
RESP	0.038	3	55	98
A.COUNSEL	0.517	23	21	22

Entity label	F1	TP	FP	FN
R.COUNSEL	0.364	16	41	15
JUDGE	0.493	17	17	18
WIT	0.000	0	11	21
AUTH	0.273	68	225	137
COURT	0.752	165	64	45
STAT	0.607	205	127	139
PREC	0.265	69	189	193
DATE	0.948	319	22	13
CASENO	0.562	81	69	57

Verification and reproducibility

The prediction ZIP has a single root prediction.json, passed CRC, and contains exactly 35 official test IDs. Its SHA-256 was compared with the completed run record. Each copied code file has a SHA-256 listed in evidence/SOURCE_HASHES.json. Exact re-execution requires the original selected weights, tokenizer and course data preserved in the complete archive; this compact package does not embed them or assert that historical notebook output cells existed.